

Shrabosty Deb

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RESEARCH PROFILE

Computer Science researcher with interests in machine learning, computer vision, predictive analytics, time-series modeling, agent-based systems, and applied data science. Published peer-reviewed research in deep learning-based image classification and interdisciplinary environmental studies. Experienced in developing computational solutions for large-scale spatiotemporal datasets, predictive modeling, simulation-based analysis, and data-driven decision-making. Current research focuses on machine learning applications, climate and environmental data analytics, streamflow prediction, groundwater sustainability modeling, and agent-based modeling of human adaptation behavior.

RESEARCH INTERESTS

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|------------------------|------------------------|--------------------------------|
| • Machine Learning | • Deep Learning | • Computer Vision |
| • Image Classification | • Predictive Analytics | • Time-Series Forecasting |
| • Data Mining | • Explainable AI | • AI for Environmental Systems |
| • Agent-Based Modeling | • Applied Data Science | |

EDUCATION

Bachelor of Science in Computer Science and Engineering | Daffodil International University, Dhaka, Bangladesh | May 2016 - January 2020

- Academic CGPA: 3.63/4.00

RESEARCH EXPERIENCE

Independent Researcher – Computer Vision, Deep Learning, Computational ML, and Environmental Data Analytics | 2021 - Present

- Developed convolutional neural network models for image classification tasks in agricultural and food image recognition.
- Contributed to two IEEE conference publications on potato and dried fish classification using deep learning.
- Worked on image preprocessing, dataset preparation, model training, evaluation, and performance analysis.
- Develop computational workflows for large-scale spatiotemporal datasets and scenario-based analysis.
- Analyze CMIP6 climate model outputs and bias-corrected projections using Python-based workflows.
- Apply predictive analytics and time-series modeling to streamflow and environmental datasets.
- Design simulation-based experiments to study household adaptation behavior and policy interventions.

PUBLICATIONS

Journal Articles

1. Rana, M. F., Ar Rafi, A., Saha, O. R., Haque, A., **Deb, S.**, Saha Sonet, S., & Islam, M. R. (2026). **Socioeconomic consequences of riverbank erosion and accretion along the Dhaleshwari River in Tangail district of Bangladesh.** Discover Geoscience, 4, Article 31. [\[Paper Link\]](#)

Conference Proceedings

1. **Deb, S.**, Laboni, M. A., & Hena, M. H. (2020). **Bangladeshi Local Potatoes Dataset and Classification Using Deep Learning.** In 2020 3rd International Conference on Intelligent Sustainable Systems (ICISS), IEEE, pp. 672-676. [\[Paper Link\]](#)
2. Laboni, M. A., Rimi, I. F., **Deb, S.**, Afrin, F., & Hena, M. H. (2022). **Dried Fish Classification Using Deep Learning.** In 2022 Second International Conference on Advances in Electrical, Computing, Communication and Sustainable Technologies (ICAECT), IEEE, pp. 1-6. [\[Paper Link\]](#)

Manuscript Under Review

- **Modeling the Operation of the Bengal Water Machine for Groundwater Sustainability in Northwestern Bangladesh.**
Author List: Ovi Ranjan Saha, **Shrabosty Deb**, Md. Ashraful Islam
Under review at Scientific Report (Springer Nature). Submitted: November 2025.
[Submitted version will be available upon request]
- **Projected Changes in Climate Extremes across Bangladesh under CMIP6 Scenarios.**
Author List: **Shrabosty Deb**, Ovi Ranjan Saha
Under review at Discover Hazards (Springer Nature). Submitted: May 2026.
[Submitted version will be available upon request]

Manuscripts in Preparation

- Streamflow Prediction Across the United States Using Advanced Machine Learning Approaches
Author List: **Shrabosty Deb**, Ovi Ranjan Saha, Abdullah Al Mehedi.
[Codes are available on GitHub]
- Agent-Based Model of Household Flood Adaptation in Bangladesh.
Author List: Shrabosty Deb, Md. Raiful Islam
[Codes are available on GitHub]

PROFESSIONAL EXPERIENCE

Product Manager | IPDC Finance PLC | 2025 - Present

- Work on product strategy, business analytics, and data-driven decision support for retail financial products.
- Analyze customer behavior and portfolio performance to support product development and service improvement.

Associate Manager, Portfolio and Data Analysis | BRAC Bank PLC | February 2022 - 2025

- Built and evaluated predictive models for credit risk scoring using real-world financial data.
- Performed feature engineering, model validation, and performance monitoring on large, noisy datasets.
- Collaborated with business and risk teams to translate analytical outputs into operational decisions.

TECHNICAL EXPERTISE

- Machine Learning and Deep Learning
- Predictive Analytics and Time-Series Modeling
- Large-Scale Data Processing
- Statistical Analysis and Data Visualization
- Computer Vision and Image Classification
- Agent-Based Modeling and Simulation
- Python-Based Scientific Computing
- Computational Modeling for Environmental Data

Programming and Tools: Python, Java, C/C++, Scikit-learn, TensorFlow, Keras, PyTorch, XGBoost, pandas, NumPy, Matplotlib, SQL, Firebase, Git, Linux, Jupyter Notebook, Google Colab, PyCharm.

TEACHING EXPERIENCE

Undergraduate Teaching Assistant | Department of Computer Science and Engineering, Daffodil International University

- Courses: Programming and Problem Solving; Algorithms.
- Assisted students in algorithm design, implementation, lab work, and programming practice.

Trainer - 120 Minutes of C | Daffodil International University

- Delivered introductory C programming training sessions for beginner-level students.

SELECTED PROJECTS

Medilink - Android-Based Healthcare Support System | Final Year Project

- Developed an Android application for healthcare appointment and service management.
- Integrated Firebase database, patient-doctor interfaces, and optimized user workflows.
- Supervisor: Mr. Sheikh Abujar, Senior Lecturer, Daffodil International University.
- Link: [Project Repository](#)

Killassin - Java-Based Game Development Project

- Developed a game-based application using object-oriented programming principles.
- Implemented game logic, user interaction mechanisms, and graphical interfaces.
- Supervisor: Prof. Dr. Syed Akhter Hossain, Daffodil International University.
- Links: [GitHub](#), [Demo Video](#)

Weight Measurement Machine - Embedded System Project

- Designed an embedded system project for weight measurement as part of academic coursework.
- Supervisor: Shams Shahriar, Lecturer, Daffodil International University.
- Link: [Project Link](#)

HONORS AND ACHIEVEMENTS

- 12th Position, National Girls' Programming Contest, 2017.
- 1st Position among all-girls teams, National Collegiate Programming Contest, 2018.
- 38th Position, NSU Inter-University Girls' Programming Contest, 2018.

LEADERSHIP EXPERIENCE

President | Daffodil International University Girls' Computer Programming Club | January 2019 - January 2020

General Secretary | Daffodil International University Girls' Computer Programming Club | January 2018 - January 2019

- Organized programming contests, workshops, seminars, and technical events, including National Girls Programming Contest and DIU programming contests.

References

Dr. Syed Akhter Hossain

Dean & Professor

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UCSI University Bangladesh Branch Campus

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Most. Hasna Hena

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